

Food Bank Resource Allocation Optimizer

RL decision-support prototype for fair, nutritious and low-waste food distribution

CT-469 Reinforcement Learning

Instructor: Dr. Murk Marvi

Group: AI 22041 Talha Noor, AI 22047 Sufiyan Nadeem, AI 22044 Ali Rana

NED University of Engineering and Technology

Problem

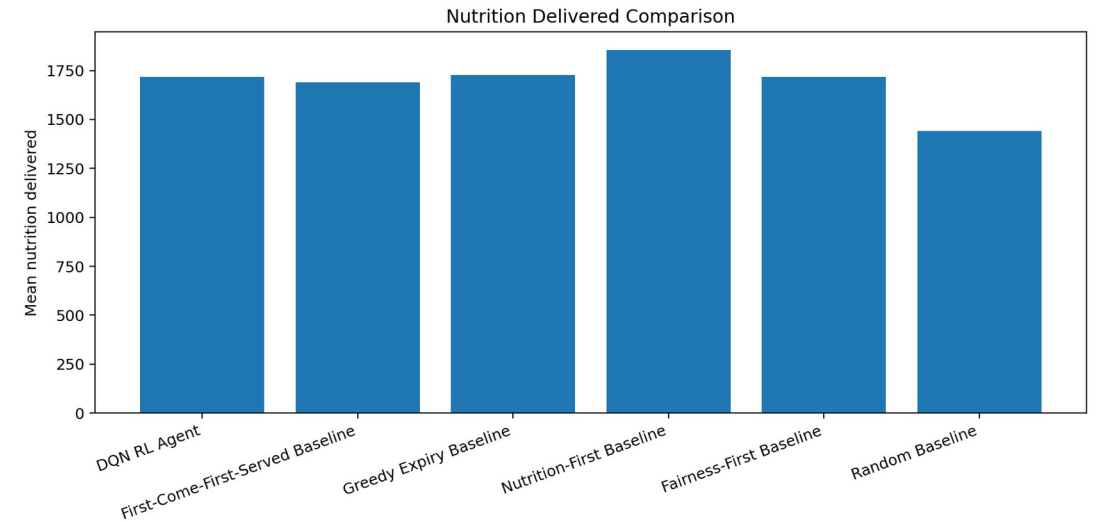
Food bank allocation is not only a queue problem

Limited donations must be allocated before expiry.

Families have different sizes, need levels and dietary restrictions.

Manual rules can maximize one metric but fail on fairness or dignity.

Goal: learn a balanced allocation policy and compare it to non-RL baselines.



Deliverables Covered

Source code repository structure with modular files.

README setup instructions.

Browser-based Streamlit web interface.

Standalone training script: train.py.

Saved DQN model: models/dqn_foodbank.pt.

Custom environment and multiple non-RL baselines.

Report and presentation with final results.

System Architecture

environment/foodbank_env.py -> custom simulation

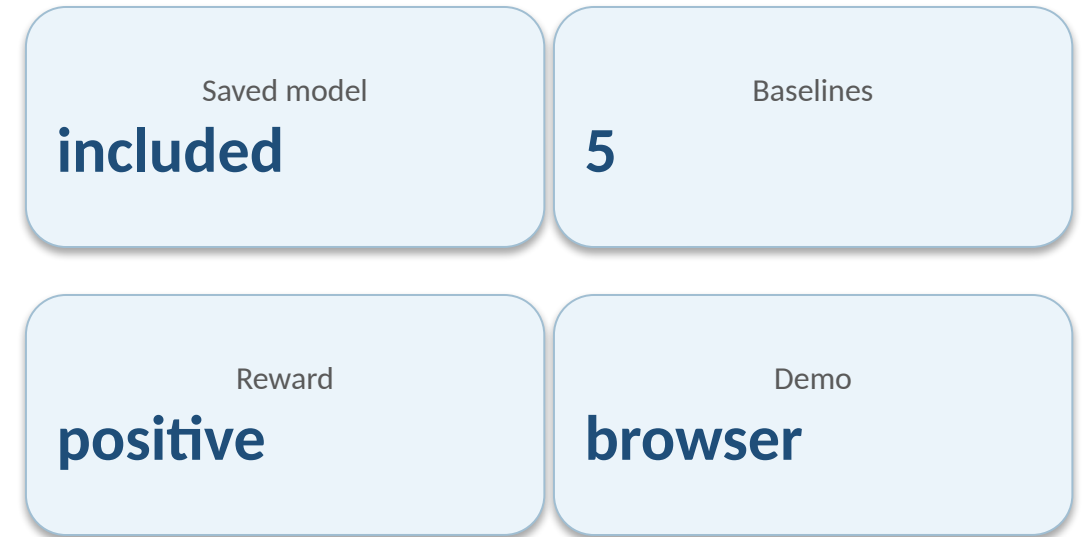
agents/dqn_agent.py -> dueling Double-DQN

agents/baselines.py -> non-RL comparison methods

train.py -> learning loop and saved model

test.py -> final evaluation graphs/CSV

app.py -> live Streamlit dashboard



RL Formulation

State: inventory quantity, days-to-expiry, nutrition, category, budget, need, coverage and fairness.

Action: choose an allocation strategy rather than a raw family ID.

Reward: positive points for useful allocation behavior.

Objective: maximize a balanced score across nutrition, fairness, waste and dietary compatibility.

DQN Agent

Dueling network: separates state value from action advantage.

Double-DQN target: reduces Q-value overestimation.

Replay buffer: learns from past allocation transitions.

Target network: improves training stability.

Evaluation uses $\epsilon=0$ for the learned greedy policy.

Non-RL Baselines

First-Come-First-Served: queue-based manual rule.

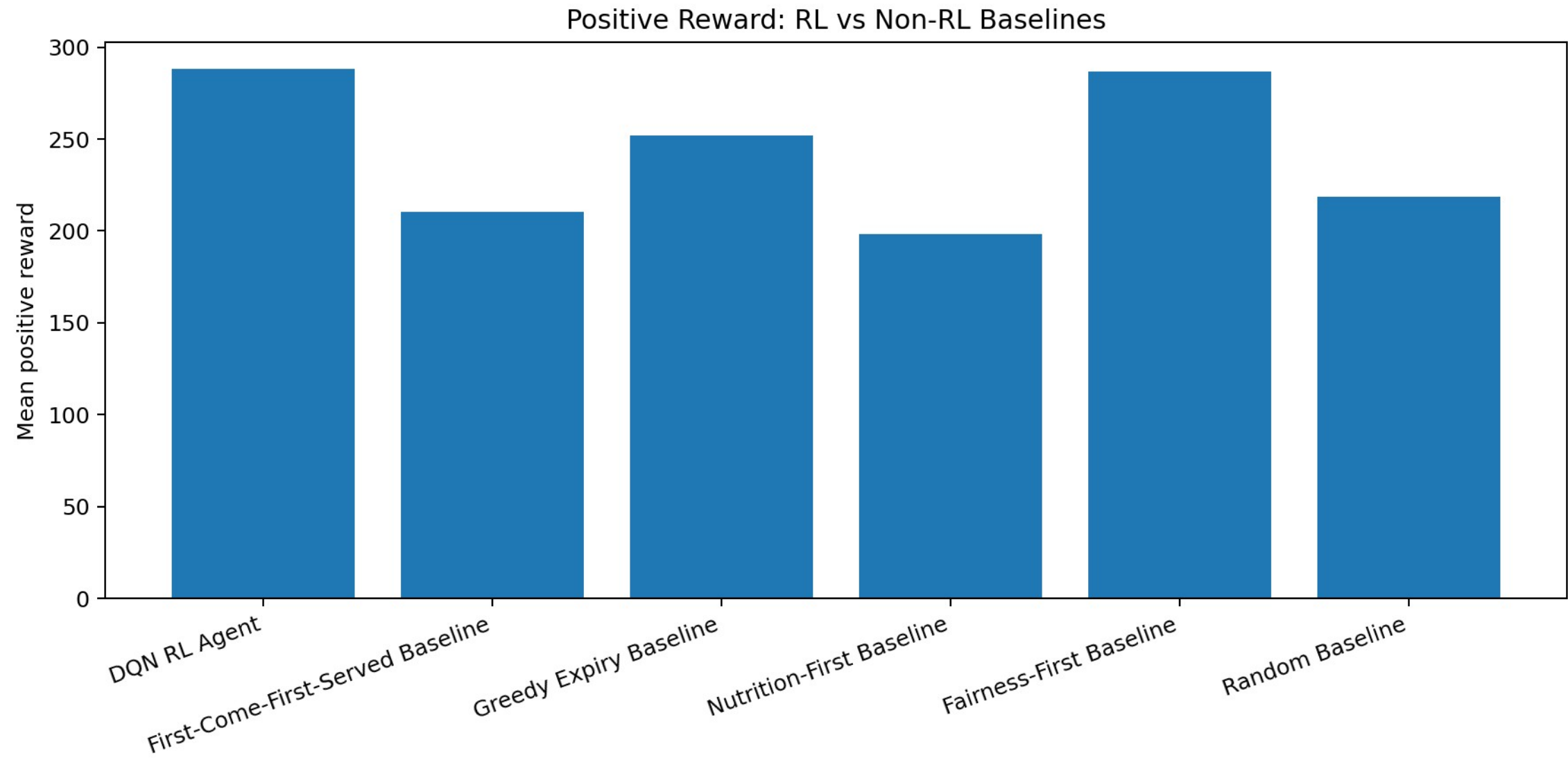
Greedy Expiry: prioritizes food closest to expiry.

Nutrition-First: maximizes raw nutrition only.

Fairness-First: prioritizes under-covered families.

Random: lower-bound reference.

Final Result: Positive Reward



Result Interpretation

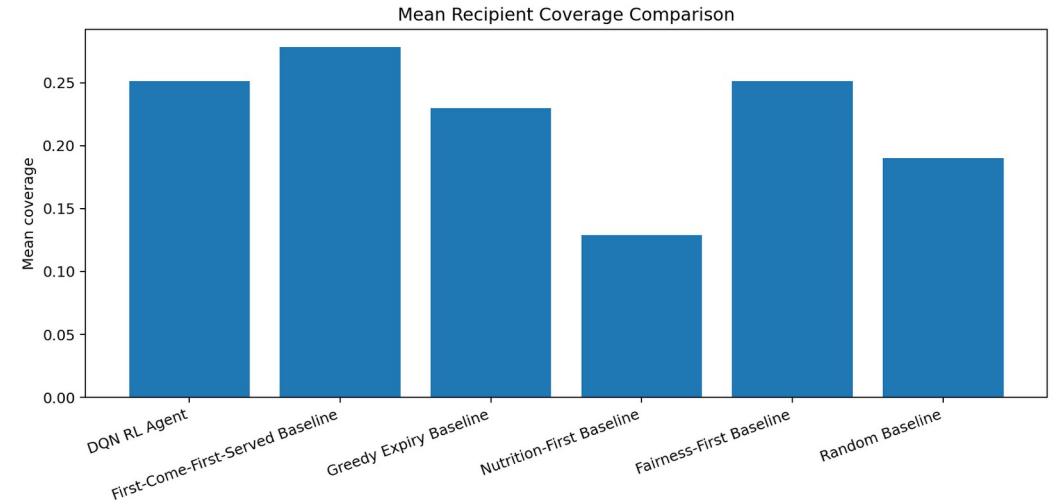
DQN achieved the highest mean reward: 288.21.

Fairness-First was the strongest non-RL competitor: 286.86.

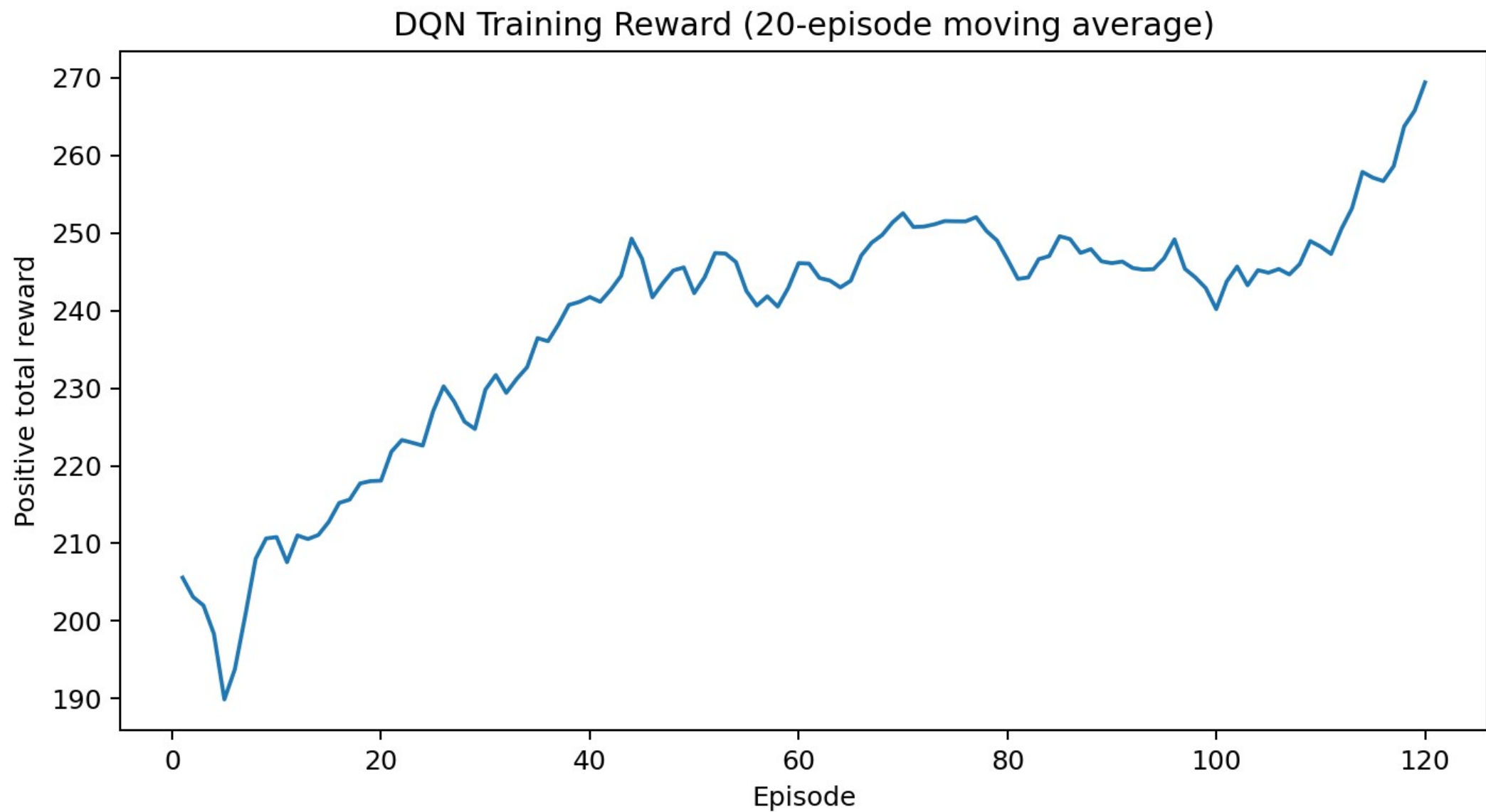
Nutrition-First delivered high raw nutrition but had weak fairness and coverage balance.

DQN reduced dietary mismatches to 2.58, far below FCFS (14.50).

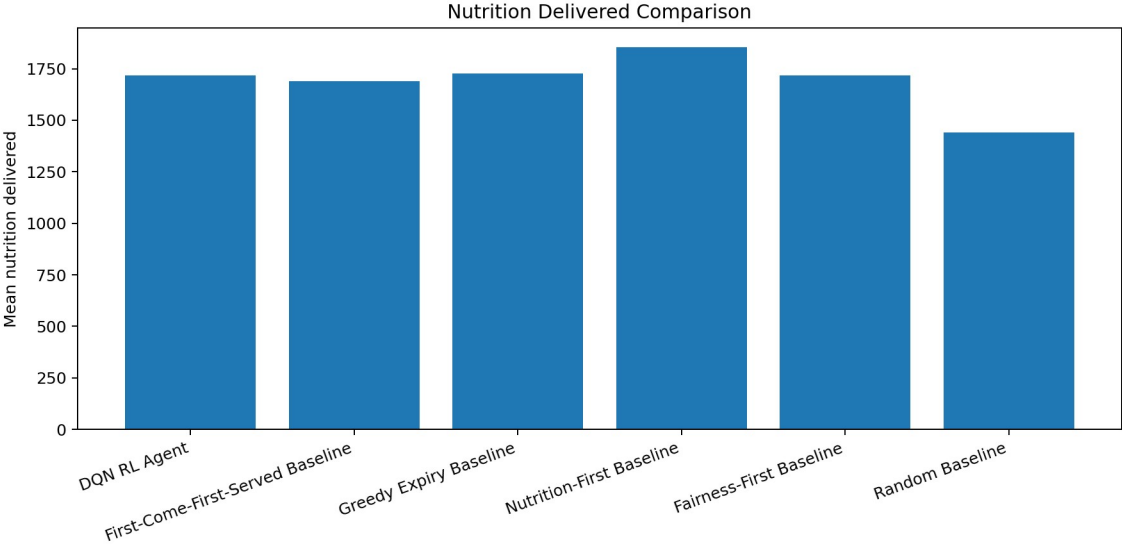
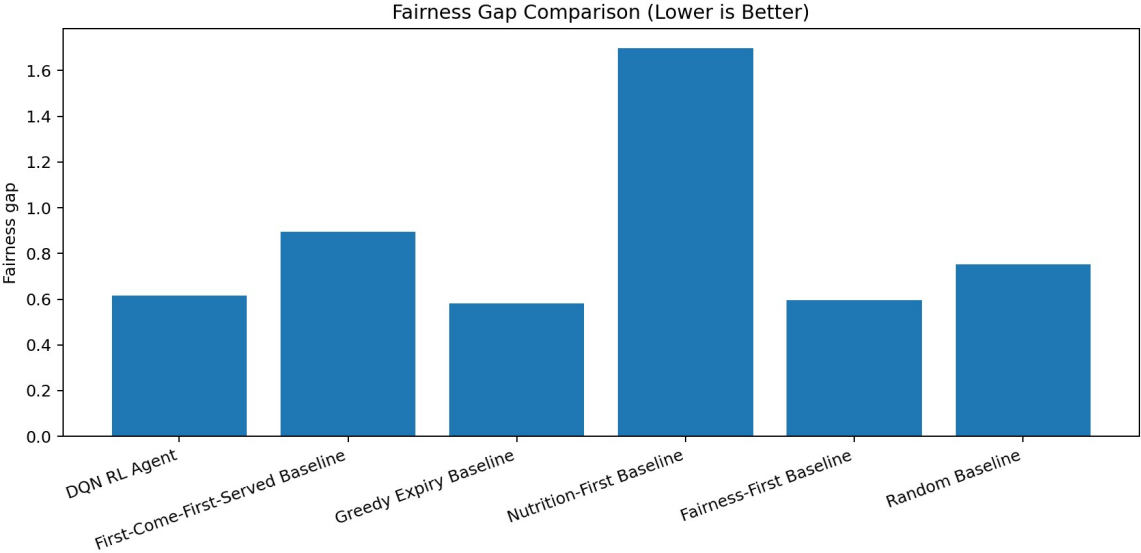
Conclusion: DQN learned the best overall multi-objective policy.



Training Progress



Fairness and Constraints



Live Demo Plan

Run: `python -m streamlit run app.py`

Show starting inventory and recipient snapshot.

Click “Run one full comparison episode”.

Explain Reward, Nutrition, Expired Units, Fairness Gap, Mean Coverage and Mismatches.

Mention that final evaluation is based on repeated `test.py` episodes.

Limitations and Future Work

Data is simulated, not collected from a real food bank.

Expiry dynamics are simplified.

Future: real donation data, multi-location routing, demand forecasting, stronger constrained RL and recipient self-selection.

Conclusion

Complete working RL prototype.

Positive, explainable reward system.

DQN trained and saved for immediate testing.

Five non-RL baselines included.

Final results show DQN has the best overall reward.